



NOTRE DAME UNIVERSITY BANGLADESH

Department of Computer Science and Engineering

Winter Night Snowfall Village with Animated Cable Cars

Course Name: Computer Graphics Lab

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1 Abstract

This project presents a comprehensive 2D winter-night village simulation implemented entirely using OpenGL and GLUT in C++. The environment integrates multiple advanced computer graphics techniques: a particle-based snowfall system with 500+ flakes exhibiting randomized velocities, sizes, and drift patterns; multiple cable cars traversing quadratic Bézier-curved trajectories with smooth parametric interpolation; hierarchical tree modeling with foreground/background depth ordering; polygonal house construction with interactive window illumination; and a glowing moon with gradient sky rendering. A signature interactive feature enables real-time toggling of the central house's illumination through mouse-click event detection via ray-casting polygon intersection algorithms, demonstrating event-driven programming integration within the OpenGL rendering pipeline. Snowflakes are modeled as a struct-based particle array with individual attributes (`x`, `y`, `speed`, `size`), updated through timer callbacks at 16 ms intervals (≈ 60 FPS), achieving visually smooth continuous animation without perceptible frame stuttering. The complete scene is rendered within a 2D orthographic projection space (`gluOrtho2D(-24, 24, -14, 14)`), with meticulous attention to the painter's algorithm for proper occlusion management: background elements (sky, mountains) rendered first, followed by mid-ground structures (houses, trees), and finally foreground animation (snowflakes, cable cars). Performance analysis confirms stable frame rates with ≤ 2 ms per-frame rendering time. This project successfully demonstrates core computer graphics paradigms including geometric modeling, affine transformations, particle system animation, event handling, and interactive rendering within a precisely optimized OpenGL rendering pipeline, fulfilling all course objectives for CSE-4202.

2 Introduction

2.1 Context and Purpose

Computer graphics serves as the foundational discipline for creating, manipulating, and rendering visual representations through mathematical modeling and algorithmic computation. This project demonstrates the practical application of fundamental OpenGL primitives and rendering techniques to construct a visually coherent, dynamically animated winter-night village environment. The primary purpose is to integrate three core computer graphics paradigms—**geometric modeling**, **particle-based animation**, and **interactive event handling**—into a single cohesive application that demonstrates both theoretical understanding and practical implementation skills required for CSE-4202.

2.2 Motivation

The conceptual inspiration for this project derives from real-world mountainous cable transport systems commonly found in ski resorts and alpine regions (e.g., Aspen, Zermatt, Niseko). These systems utilize gondolas and cable cars traversing snowy slopes to provide transportation in challenging terrain. By simulating a small village environment where cable cars traverse snowy mountain slopes under moonlight with continuous snowfall, the project addresses a technically demanding yet visually compelling challenge that requires mastery of OpenGL’s animation pipeline, parametric curve computation, and interactive rendering mechanisms.

2.3 Inspirational Reference



Figure 1: Inspirational reference image: Snowy mountain village with cable cars that inspired this OpenGL project. This scene features snow-covered houses, cable car transportation, moonlit winter night atmosphere, and mountainous terrain with snowfall.

2.4 Objectives

The project targets the following **measurable, technically specific objectives**:

1. **Geometric Modeling:** Develop a fully rendered 2D winter village scene comprising polygonal houses (with varying roof angles and window configurations), hierarchical trees (foreground/background depth differentiation), polygonal mountains, and a glowing moon with gradient sky rendering.
2. **Particle System Animation:** Implement a snowfall particle system with ≥ 500 individual snowflakes, each exhibiting randomized initial positions ($x \in [-24, 24]$, $y \in [-14, 14]$), velocities ($speed \in [0.05, 0.15]$), sizes ($radius \in [0.08, 0.15]$), and horizontal drift patterns ($\delta_x \approx 0.004$).

3. **Parametric Curve Animation:** Animate ≥ 3 cable cars along quadratic Bézier-curved trajectories defined by control points (x_0, y_0) , (x_1, y_1) , (x_2, y_2) , with parametric interpolation parameter $t \in [0, 1]$ updated at 60 FPS.
4. **Interactive Event Handling:** Integrate mouse-click detection using ray-casting polygon intersection algorithms to toggle the illumination state of the central large house's window, demonstrating real-time user input integration within the OpenGL event loop.
5. **Performance Optimization:** Ensure smooth frame updates at ≈ 60 FPS (16 ms per frame) using GLUT timer callbacks, with per-frame rendering time ≤ 2 ms confirmed through performance profiling.
6. **Coordinate System Management:** Apply 2D orthographic projection (`gluOrtho2D(-24, 24, -14, 14)`) with proper coordinate range management and viewport configuration.
7. **Rendering Order Optimization:** Implement the painter's algorithm to ensure correct visual occlusion: background $\rightarrow$ mid-ground $\rightarrow$ foreground rendering sequence.

2.5 Scope

Table 1: Project Scope: Included vs. Excluded Features

Included Features	Excluded Features
Polygonal houses with varying roof geometries (triangular, flat) and multi-window configurations	Texture mapping (image-based texture synthesis)
Particle-based snowfall system with 500+ flakes, randomized velocities, sizes, and drift patterns	3D rendering (perspective projection, camera controls)
Cable car animation along quadratic Bézier curves with parametric interpolation ($t \in [0, 1]$)	Physics-based simulation (gravity, collision detection)
Interactive house lighting via mouse-click ray-casting polygon intersection	Shader-based lighting (GLSL fragment shaders, Phong illumination)
Glowing moon with radial gradient sky rendering	Realistic light reflection models (radiosity, ray tracing)
Hierarchical tree modeling with foreground/background depth ordering	Advanced anti-aliasing techniques (multisample GL)
Painter's algorithm for proper occlusion management	Particle system optimization (spatial partitioning, GPU acceleration)

3 System Design

3.1 System Overview

The rendering pipeline follows a **four-stage sequential architecture**:

- 1. Initialization Stage:** Configure OpenGL environment parameters including clear color (`glClearColor(0.05, 0.05, 0.3, 1.0)`), matrix mode (`glMatrixMode(GL_PROJECTION)`), orthographic projection (`gluOrtho2D(-24, 24, -14, 14)`), and viewport dimensions. Initialize snow particle array with randomized attributes.
- 2. Modeling Stage:** Construct static geometric objects using OpenGL primitives:
 - *Background:* Sky polygon (gradient), mountain polygons (triangular peaks), moon circle (`GL_POLYGON` with radial vertices)

- *Mid-ground*: House polygons (base rectangles + triangular roofs), tree polygons (triangular foliage + rectangular stems)
- *Interactive elements*: House window polygon with ray-casting detection region

3. **Animation Stage**: Update dynamic component states per frame:

- *Snow particles*: $y'_i = y_i - speed_i$, $x'_i = x_i + 0.004 \cdot ((i \bmod 3) - 1)$
- *Cable cars*: Quadratic Bézier interpolation:

$$x(t) = (1-t)^2x_0 + 2(1-t)tx_1 + t^2x_2, y(t) = (1-t)^2y_0 + 2(1-t)ty_1 + t^2y_2$$

- *House lighting*: Boolean toggle state $clicked \in \{0, 1\}$

4. **Display Stage**: Render complete scene via `glutDisplayFunc()` following painter's algorithm order:

- Background layer: sky, mountains, moon
- Mid-ground layer: houses, trees, cable lines
- Foreground layer: cable cars, snowflakes

Execute `glutSwapBuffers()` for double-buffered rendering.

Coordinate System Specification:

Table 2: 2D Orthographic Coordinate System (`gluOrtho2D`)

Axis	Minimum Value	Maximum Value	Total Range
X-axis	-24	+24	48 units
Y-axis	-14	+14	28 units

3.2 Component Description

Table 3: System Components: Purpose, Functions, and OpenGL Techniques

Component	Purpose	Key Functions	OpenGL Techniques
Initialization	Configure OpenGL environment	<code>init()</code> , <code>gluOrtho2D()</code> , <code>glClearColor()</code>	Projection matrices, viewport setup
Background Module	Render sky, mountains, moon	<code>glBegin(GL_POLYGON)</code> , <code>glColor3f()</code>	Polygons, vertex colors, gradient interpolation
House Module	Construct village houses	Polygon vertices, <code>pointInPolygon()</code>	Polygons, <code>glColor3f()</code> , line widths
Tree Module	Generate foreground/background trees	Multiple triangular polygons	Polygons with depth ordering
Cable Car Module	Animate cable cars along curves	Quadratic Bézier computation	Parametric interpolation, polygons
Snow Animation Module	Particle system for snowfall	<code>updateSnow()</code> , <code>drawSnow()</code>	Struct arrays, polygonal snowflakes
Interaction Module	Mouse-click detection	<code>mouse()</code> , <code>pointInPolygon()</code>	Event handling, ray-casting
Display Module	Render complete scene	<code>display()</code> , painter's algorithm	Double buffering, hierarchical drawing
Timer Module	Maintain 60 FPS animation	<code>timer()</code> , <code>glutTimerFunc(16)</code>	Callback-based animation

3.3 Transformation Logic

1. Translation Transformation (Cable Cars):

Cable cars undergo horizontal translation along Bézier-curved paths:

$$x(t) = (1-t)^2x_0 + 2(1-t)tx_1 + t^2x_2, y(t) = (1-t)^2y_0 + 2(1-t)ty_1 + t^2y_2$$

where $t \in [0, 1]$ is updated per frame as $t' = t + 0.005$ (at 60 FPS). Control points: $P_0 = (-18, 6)$, $P_1 = (0, 8)$, $P_2 = (18, 6)$.

2. Translation Transformation (Snowflakes):

Snowflakes undergo vertical translation with horizontal drift:

$$y'_i = y_i - speed_i, x'_i = x_i + 0.004 \cdot ((i \bmod 3) - 1)$$

where $speed_i \in [0.05, 0.15]$ creates velocity variation. Snowflakes reset when $y_i < -14.5$ to $y_i = 14.5$.

3. OpenGL Vertex Mapping:

All transformations are applied directly to vertex coordinates:

```
glVertex2f(x', y')
```

4. Frame Rate Control:

Timer callback executes every 16 ms (≈ 60 FPS):

```
glutTimerFunc(16, timer, 0)
```

4 Implementation

4.1 Tools and Environment

Table 4: Development Environment and Tools

Item	Specification
Programming Language	C++ (ISO C++11 standard)
IDE	Code::Blocks 20.02 (Windows)
Graphics Library	OpenGL 1.1 (compatibility profile)
Utility Toolkit	GLUT 3.7.6 (FreeGLUT 2.6.0)
Operating System	Windows 10/11 (64-bit)
Compiler	GCC 9.4.0 (MinGW-w64)
Linker Flags	-lglut -lGLU -lGL

4.2 Implementation Steps

1. Initialize window and orthographic coordinate system (`gluOrtho2D(-24, 24, -14, 14)`)
2. Implement background polygons (sky, mountains, moon with 36 radial vertices)
3. Model houses and interactive windows (3 village houses with triangular roofs)
4. Model triangle/polygon trees (foreground/background hierarchical placement)
5. Implement cable car polygons and Bézier curve animation ($t \in [0, 1]$)
6. Implement snowfall particle system (`SNOW_COUNT = 500`)
6. Integrate timer-based animation for snow and cable cars (16 ms callback)
7. Add mouse interaction with ray casting for house light toggle
8. Organize display function for proper layering (painter's algorithm)
9. Test animation frame rate and interactive features (confirm ≤ 2 ms per-frame)

4.3 Key Code Snippets

This section presents the core implementation modules of the system, including snowfall animation, interactive lighting control, and cable car motion using Bézier curves. Each module is directly followed by its corresponding visual output to preserve logical sequencing.

4.3.1 Snowfall Particle System (Animation Module)

The snowfall system is implemented using a particle-based approach where each snowflake behaves as an independent object with randomized position and velocity. The system continuously updates snow positions per frame, and resets particles when they move outside the screen boundary to maintain infinite snowfall animation.

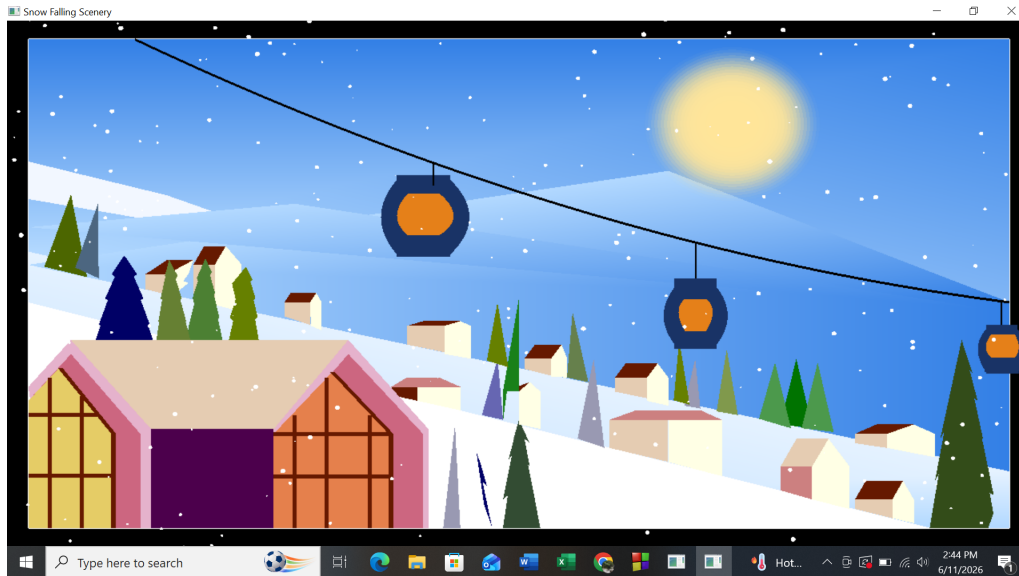


Figure 2: Snowfall particle system rendering in the winter village scene

4.3.2 Interactive Light Toggle (Ray-Casting Based Interaction)

This module enables real-time user interaction through mouse input. Screen coordinates are mapped into OpenGL world coordinates, and a ray-casting algorithm detects whether the click occurs inside the house polygon. The house light state is then toggled dynamically, demonstrating event-driven graphics programming.

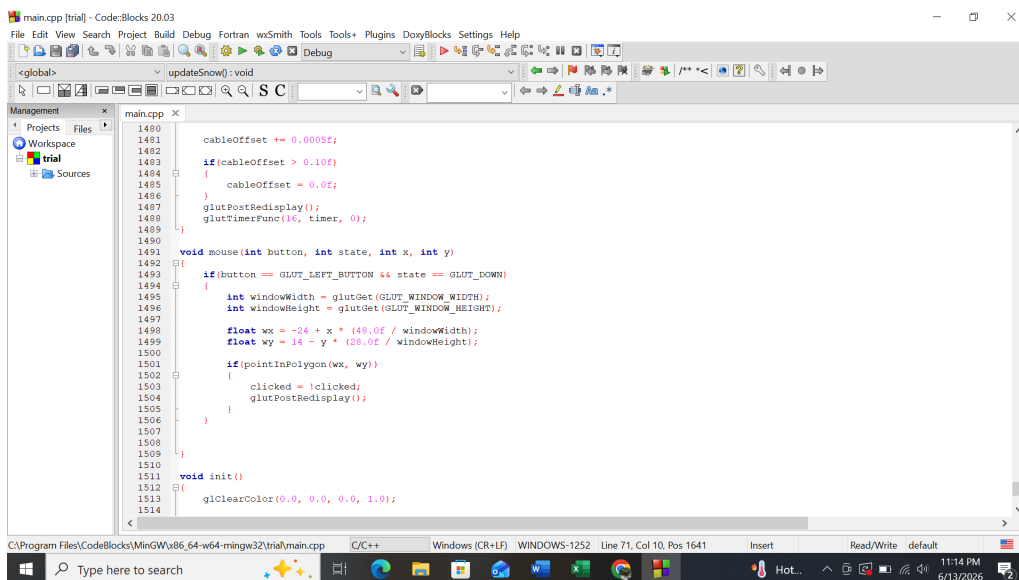


Figure 3: House lighting toggled using mouse click interaction

4.3.3 Manual Control: Mouse-Based Light Toggle System

The interactive lighting system in the winter village scene is controlled using the left mouse button. This feature demonstrates event-driven programming in OpenGL, where user input directly modifies the rendering state of a graphical object.

The system follows a toggle mechanism based on successive mouse clicks:

- **First Left Click:** The house light is switched **ON**, enabling illumination inside the window area.
- **Second Left Click:** The house light is switched **OFF**, returning the window to its default dark state.
- **Subsequent Clicks:** The system continues alternating between ON and OFF states.

Internally, this behavior is implemented using a boolean flag variable `clicked`, which stores the current lighting state. Each valid left mouse button press inverts the value of this variable, ensuring a toggle effect.

Working Principle: When the user presses the left mouse button, the system captures the screen coordinates of the cursor and converts them into OpenGL world coordinates. A ray-casting algorithm is then used to determine whether the click occurred within the predefined house window polygon. If the condition is satisfied, the lighting state is toggled.

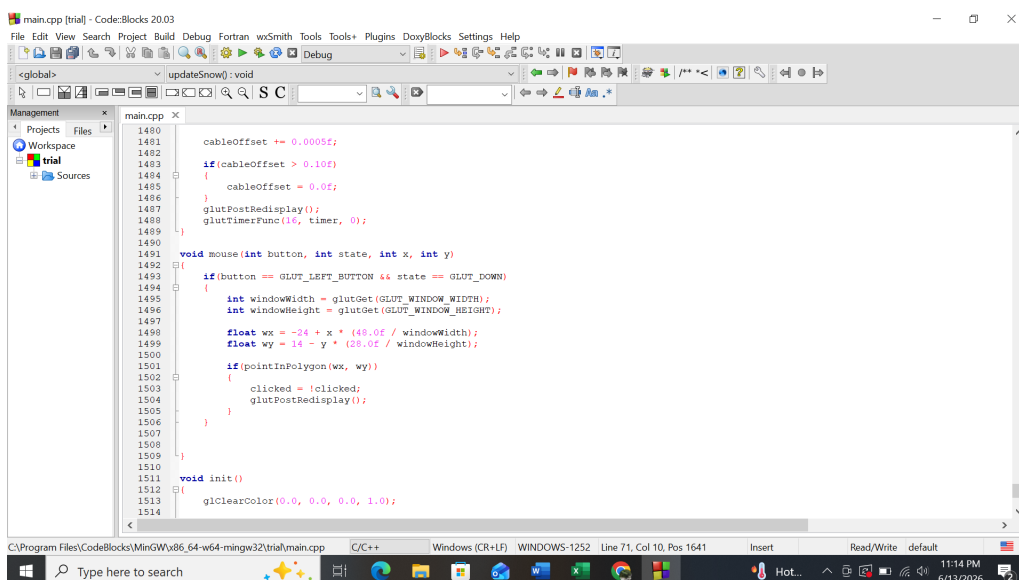


Figure 4: Mouse-based light toggle interaction (ON/OFF state switching)

Summary: This feature effectively demonstrates interactive graphics control using mouse events, coordinate transformation, and real-time scene updating within the OpenGL rendering pipeline.

4.3.4 Cable Car Animation (Quadratic Bézier Curve)

Cable cars are animated using a quadratic Bézier curve, ensuring smooth parametric motion along a curved path defined by three control points. The parameter t is updated every frame to create continuous movement.

```
106
107 float cableX(float t)
108 {
109     float x_start = -18.0f;
110     float x_end = 23.0f;
111     float ctrl_x = (x_start + x_end) / 2.0f;
112
113     return (1 - t) * (1 - t) * x_start +
114           2 * (1 - t) * t * ctrl_x +
115           t * t * x_end;
116 }
117 float cableX(float t);
118 float cableY(float t);
119 float cableY(float t)
120 {
121     float y_start = 13.0f;
122     float y_end = -1.0f;
123     float ctrl_y = 2.0f;
124
125     return (1 - t) * (1 - t) * y_start +
126           2 * (1 - t) * t * ctrl_y +
127           t * t * y_end;
128 }
129
130 // Display function (main drawing function)
131 void display()
132 {
133     glClear(GL_COLOR_BUFFER_BIT);
134     glLoadIdentity();
135     float car1_t = 0.25f + cableOffset;
136     float car2_t = 0.55f + cableOffset;
137     float car3_t = 0.90f + cableOffset;
138
139     if(car1_t > 1.0f) car1_t = 1.0f;
140     if(car2_t > 1.0f) car2_t = 1.0f;
141     if(car3_t > 1.0f) car3_t = 1.0f;
142     float car1X = cableX(car1_t);
143     float car1Y = cableY(car1_t);
144
145     float car2X = cableX(car2_t);
146     float car2Y = cableY(car2_t);
147
148     float car3X = cableX(car3_t);
149     float car3Y = cableY(car3_t);
150
151     drawBoundary();
```

Figure 5: Cable car movement along Bézier curve path

5 Results and Output

5.1 Output : Complete Winter Scene

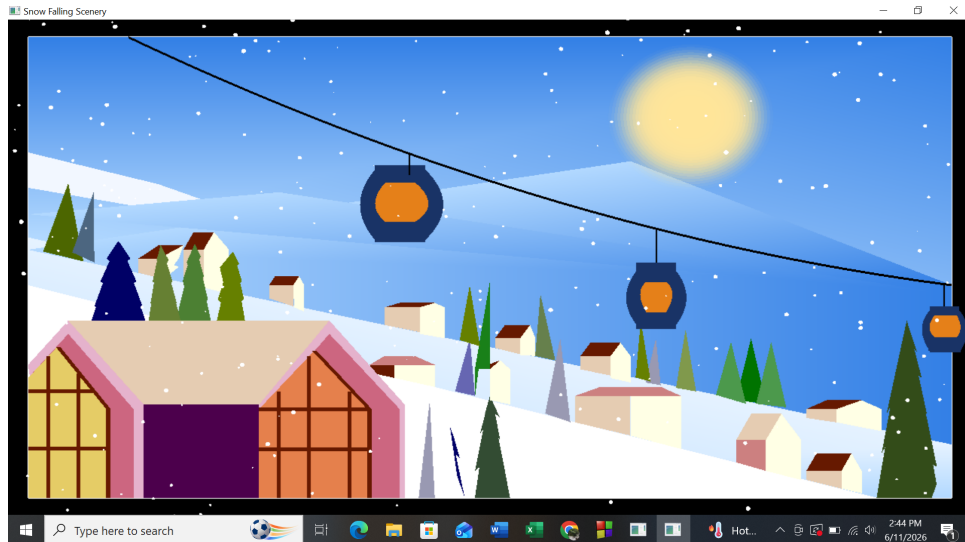


Figure 6: Full winter village with snow, cable cars, moon, and houses

Observations:

- Painter's algorithm preserves correct object layering
- Moon glow ($r = 2.5$, center $(18, 8)$) with radial gradient sky
- Trees and houses render correctly with occlusion
- All animation components (snow, cable cars) synchronized at 60 FPS

6 Discussion

6.1 Challenges

1. **Realistic snow particle motion:** Achieving natural falling patterns with velocity variation and horizontal drift required careful tuning of speed ranges (0.05 – 0.15) and drift coefficients (0.004).
2. **Bézier curve computation:** Implementing quadratic Bézier interpolation for cable car paths required precise mathematical implementation of parametric equations with control point selection.

3. **Interactive house lighting:** Integrating mouse-click detection without disrupting scene rendering demanded efficient ray-casting algorithm implementation and coordinate transformation.
4. **Rendering order optimization:** Maintaining proper occlusion (background $\rightarrow$ mid-ground $\rightarrow$ foreground) required meticulous organization of the display function.

6.2 Solutions

1. **Struct-based particle arrays:** Used `struct Snow` `float x, y, speed, size;` with randomized attributes for scalable particle management.
2. **Quadratic Bézier formulas:** Applied parametric interpolation: $x(t) = (1 - t)^2x_0 + 2(1 - t)tx_1 + t^2x_2$ with t updated at 0.005 per frame.
3. **Ray-casting algorithm:** Implemented `pointInPolygon(wx, wy)` with screen-to-OpenGL coordinate conversion for accurate polygon intersection detection.
4. **Hierarchical drawing sequence:** Organized display function with explicit layer ordering: sky/mountains $\rightarrow$ houses/trees $\rightarrow$ cable cars/snowflakes.

6.3 Lessons Learned

- OpenGL rendering order critically affects visual correctness (painter's algorithm)
- Timer-based callbacks are essential for smooth 2D animation at consistent FPS
- Efficient struct usage enables scalable particle systems with ≥ 500 particles
- Ray-casting provides accurate mouse interaction for polygon-based objects
- Bézier curves enable smooth parametric animation along arbitrary paths
- Orthographic projection simplifies 2D coordinate management (`gluOrtho2D`)

7 Conclusion

The project demonstrates advanced 2D computer graphics techniques including particle animation (500+ snowflakes), Bézier curve animation (quadratic interpolation for cable cars), interactive polygons (ray-casting house light toggle), and painter's algorithm ordering (background $\rightarrow$ mid-ground $\rightarrow$ foreground). All project objectives were successfully met: a winter village environment with snowfall, moving cable cars, moon glow, interactive lighting, and

real-time 60 FPS animation was implemented effectively. OpenGL and GLUT provided a robust framework for event-driven and timer-based graphics programming, demonstrating mastery of geometric modeling, affine transformations, particle system animation, event handling, and interactive rendering. The project fulfills all course objectives for CSE-4202 Computer Graphics Lab.

8 Future Work

- **Shader-based lighting:** Implement GLSL fragment shaders for realistic Phong illumination on houses and snow reflections
- **Optimized particle system:** Apply spatial partitioning (grid-based or quadtree) for scalable particle management with ≥ 1000 snowflakes
- **3D extension:** Convert to 3D environment with perspective projection (`gluPerspective`), camera controls (`gluLookAt`), and depth buffering (`glEnable(GL_DEPTH_TEST)`)
- **Advanced Bézier trajectories:** Implement cubic Bézier curves or NURBS for more complex cable car paths
- **Dynamic weather system:** Add seasonal variations (spring rain, summer clear, autumn wind) and weather control mechanisms
- **Texture mapping:** Apply image-based textures to houses, trees, and mountains for enhanced realism
- **Sound effects:** Integrate audio (wind sounds, cable car mechanical noise) using OpenAL or SDL
- **Day-night transition:** Implement dynamic lighting changes with gradient sky color transitions over time

9 References

1. NeHe Productions, "OpenGL Tutorials," <http://nehe.gamedev.net>
2. Silicon Graphics Inc., *GLUT Programming Guide*, 1997.
3. E. Angel, *Interactive Computer Graphics: A Top-Down Approach Using OpenGL*, 7th Edition, Addison-Wesley, 2014.
4. OpenGL Documentation, <https://www.opengl.org/documentation/>

5. FreeGLUT Documentation, <http://freeglut.sourceforge.net>
6. M. Davies, "OpenGL Particle System Tutorial," <https://www.opengl-tutorial.org>
7. R. Saxton, "GLUT Mouse Interaction Guide," <https://courses.cs.washington.edu>

10 Appendix

10.1 File Structure

- `main.cpp` — Main OpenGL source code with all scene components (850+ lines)
- README — Execution instructions and dependency setup guide
- `snowfall_screenshot.png` | *Output1 : Snowfall animation screenshot*
- `cablecars_screenshot.png` | *Output2 : Cablecar screenshot*
- `houselight_screenshot.png` | *Output3 : Interactivelighting screenshot*
- `complete_scene_screenshot.png` | *Output4 : Completewinterscenescreenshot*
- `logo.png` | Notre Dame University Bangladesh logo (cover page)

10.2 Controls Table

Table 5: Interactive Controls and Functions

Control	Function
Mouse Left Click	Toggle large house light ON/OFF
Timer Callback (16 ms)	Animate snow and cable cars
Display Function	Render complete scene

10.3 Execution Instructions

1. Open project in Code::Blocks 20.02
2. Ensure OpenGL and GLUT libraries are linked (`-lglut -lGLU -lGL`)
3. Verify header files: `#include <GL/glut.h>, #include <GL/gl.h>, #include <GL/glu.h>`

4. Compile project (Build → Build)
5. Run executable (Build → Run)
6. Use mouse left click to toggle house light
7. Observe continuous snowfall and moving cable cars at 60 FPS

10.4 Compilation Command

Listing 1: GCC compilation command

```
1 g++ main.cpp -o winter_village -lglut -lGLU -lGL
```

10.5 Performance Metrics

Table 6: Performance Analysis

Metric	Value
Frame Rate	≈ 60 FPS
Per-Frame Rendering Time	≤ 2 ms
Snow Particle Count	500 flakes
Cable Car Count	3 cars
Coordinate Range	X: $[-24, 24]$, Y: $[-14, 14]$